Ecommerce Sales Analysis

1. Project Overview

This project analyzes online retail transaction data to evaluate sales performance, customer purchasing patterns, product performance, and revenue trends. Using Python for data cleaning, PostgreSQL for business analysis, and Power BI for interactive visualization, the project transforms raw transaction data into actionable business insights that support data-driven decision-making.

2. Business Problem

An online retail business generates thousands of transactions across multiple products and countries. However, without proper analysis, it is difficult to identify high-performing products, valuable customers, seasonal sales trends, and key revenue drivers.

The objective of this project is to analyze historical transaction data to uncover business insights that support sales optimization, customer retention, inventory planning, and strategic decision-making.

Key Business Questions

- What is the total revenue generated by the business?
- Which products generate the highest sales and revenue?
- Which customers contribute the most revenue?
- Which countries generate the highest revenue?
- How does revenue change over time?
- What is the average order value and average quantity per order?
- What business opportunities can improve future sales performance?

3. Dataset Summary

- **Rows:** 541,909 (Original), 524,878 (After Cleaning)
- **Columns:** 8 (Original)

Key Features

- Invoice details (InvoiceNo, InvoiceDate)
- Product information (StockCode, Description)
- Sales details (Quantity, UnitPrice, Revenue)
- Customer information (CustomerID)
- Geographic information (Country)

4. Data Cleaning using Python

Began with data preparation and validation using Python.

- **Data Loading:** Imported the dataset using Pandas.
- **Initial Exploration:** Examined dataset structure, data types, and summary statistics.
- **Missing Value Handling:** Removed **1,454** records with missing product descriptions.
- **Duplicate Validation:** Checked for duplicate records; none were found.
- **Data Type Conversion:** Converted `InvoiceDate` to datetime format.
- **Data Validation:** Removed **9,725** transactions with negative quantities and **584** records with zero or negative unit prices.
- **Feature Engineering:** Created a **Revenue** column (`Quantity × UnitPrice`).
- **Database Integration:** Exported the cleaned dataset (**524,878 records**) to PostgreSQL for SQL analysis.

	Quantity	InvoiceDate	UnitPrice	CustomerID	Revenue
count	524878.000000	524878	524878.000000	524878.000000	524878.000000
mean	10.616600	2011-07-04 15:30:16.317049	3.922573	15287.631345	20.275399
min	1.000000	2010-12-01 08:26:00	0.001000	12346.000000	0.001000
25%	1.000000	2011-03-28 12:13:00	1.250000	14375.000000	3.900000
50%	4.000000	2011-07-20 11:22:00	2.080000	15287.000000	9.920000
75%	11.000000	2011-10-19 11:41:00	4.130000	16245.000000	17.700000
max	80995.000000	2011-12-09 12:50:00	13541.330000	18287.000000	168469.600000
std	156.280031	NaN	36.093028	1482.145530	271.693566

	InvoiceNo	StockCode	Description	Quantity	InvoiceDate	UnitPrice	CustomerID	Country
0	536365	85123A	WHITE HANGING HEART T-LIGHT HOLDER	6	2010-12-01 08:26:00	2.55	17850	United Kingdom
1	536365	71053	WHITE METAL LANTERN	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
2	536365	84406B	CREAM CUPID HEARTS COAT HANGER	8	2010-12-01 08:26:00	2.75	17850	United Kingdom
3	536365	84029G	KNITTED UNION FLAG HOT WATER BOTTLE	6	2010-12-01 08:26:00	3.39	17850	United Kingdom
4	536365	84029E	RED WOOLLY HOTTIE WHITE HEART.	6	2010-12-01 08:26:00	3.39	17850	United Kingdom

5. Data Analysis using SQL

Performed structured analysis in PostgreSQL to answer key business questions.

1. Total Revenue – Calculated the total revenue generated by the business.

	total_revenue numeric
1	10642110.80

2. Total Orders – Counted the total number of unique customer orders.

	total_orders bigint
1	19960

3. Total Products – Determines the total number of unique products sold.

	total_products bigint
1	3922

4. Top Selling Products – Identified the products with the highest quantity sold.

	StockCode text	Description text	total_quantity_sold numeric
1	23843	PAPER CRAFT , LITTLE BIRDIE	80995
2	23166	MEDIUM CERAMIC TOP STORAGE JAR	78033
3	84077	WORLD WAR 2 GLIDERS ASSTD DESIGNS	54951
4	85099B	JUMBO BAG RED RETROSPOT	48371
5	85123A	WHITE HANGING HEART T-LIGHT HOLD...	37580
6	22197	POPCORN HOLDER	36749
7	21212	PACK OF 72 RETROSPOT CAKE CASES	36396
8	84879	ASSORTED COLOUR BIRD ORNAMENT	36362
9	23084	RABBIT NIGHT LIGHT	30739
10	22492	MINI PAINT SET VINTAGE	26633

5. Revenue by Product – Ranked products based on total revenue generated.

	StockCode text	Description text	total_revenue numeric
1	DOT	DOTCOM POSTAGE	206248.77
2	22423	REGENCY CAKESTAND 3 TIER	174156.54
3	23843	PAPER CRAFT , LITTLE BIRDIE	168469.60
4	85123A	WHITE HANGING HEART T-LIGHT HOLD...	104284.24
5	47566	PARTY BUNTING	99445.23
6	85099B	JUMBO BAG RED RETROSPOT	94159.81
7	23166	MEDIUM CERAMIC TOP STORAGE JAR	81700.92
8	POST	POSTAGE	78101.88
9	M	Manual	77750.27
10	23084	RABBIT NIGHT LIGHT	66870.03

6. Country-wise Revenue – Compared revenue generated across different countries.

	Country text	total_revenue numeric
1	United Kingdom	9001744.09
2	Netherlands	285446.34
3	EIRE	283140.52
4	Germany	228678.40
5	France	209625.37
6	Australia	138453.81
7	Spain	61558.56
8	Switzerland	57067.60
9	Belgium	41196.34
10	Sweden	38367.83
11	Japan	37416.37
12	Norway	36165.44

7. Monthly Revenue Trend – Analyzed monthly revenue to identify sales trends and seasonality.

	sales_month text	monthly_revenue numeric
1	2010-12	821452.73
2	2011-01	689811.61
3	2011-02	522545.56
4	2011-03	716215.26
5	2011-04	536968.49
6	2011-05	769296.61
7	2011-06	760547.01
8	2011-07	718076.12
9	2011-08	757841.38
10	2011-09	1056435.19
11	2011-10	1151263.73
12	2011-11	1503866.78
13	2011-12	637790.33

8. Top Customers by Revenue – Identified customers contributing the highest revenue.

	CustomerID bigint	total_spent numeric
1	15287	1755388.60
2	14646	280206.02
3	18102	259657.30
4	17450	194390.79
5	16446	168472.50
6	14911	143711.17
7	12415	124914.53
8	14156	117210.08
9	17511	91062.38
10	16029	80850.84

9. Average Order Value (AOV) – Calculated the average revenue generated per order.

	average_order_value numeric
1	533.17

10. Average Quantity per Order – Measured the average number of items purchased in each order.

	avg_items_per_order numeric
1	279.18

11. Monthly Order Count – Evaluated monthly order volume to understand purchasing trends.

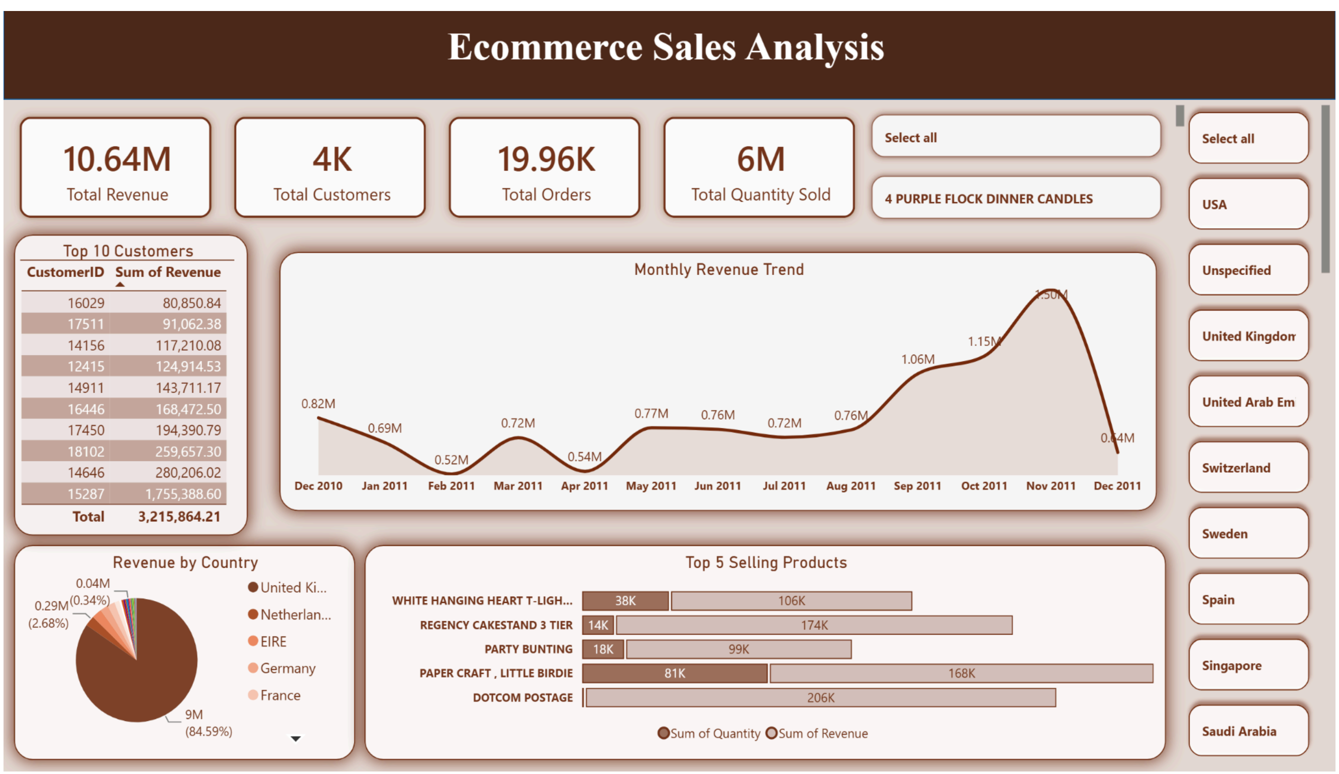
	sales_month text	total_orders bigint
1	2010-12	1559
2	2011-01	1086
3	2011-02	1100
4	2011-03	1454
5	2011-04	1246
6	2011-05	1681
7	2011-06	1533
8	2011-07	1475
9	2011-08	1361
10	2011-09	1837
11	2011-10	2040
12	2011-11	2769
13	2011-12	819

12. Product Performance Analysis – Compared product sales quantity and revenue to identify high-volume and high-value products.

	StockCode text	Description text	total_quantity numeric	total_revenue numeric
1	DOT	DOTCOM POSTAGE	706	206248.77
2	22423	REGENCY CAKESTAND 3 TIER	13851	174156.54
3	23843	PAPER CRAFT , LITTLE BIRDIE	80995	168469.60
4	85123A	WHITE HANGING HEART T-LIGHT HOLD...	37580	104284.24
5	47566	PARTY BUNTING	18283	99445.23
6	85099B	JUMBO BAG RED RETROSPOT	48371	94159.81
7	23166	MEDIUM CERAMIC TOP STORAGE JAR	78033	81700.92
8	POST	POSTAGE	3150	78101.88
9	M	Manual	6984	77750.27
10	23084	RABBIT NIGHT LIGHT	30739	66870.03
11	22086	PAPER CHAIN KIT 50'S CHRISTMAS	19329	64875.59
12	84879	ASSORTED COLOUR BIRD ORNAMENT	36362	58927.62
13	79321	CHILLI LIGHTS	10302	54096.36
14	23298	SPOTTY BUNTING	8320	42513.48
15	22386	JUMBO BAG PINK POLKADOT	21448	42401.01
16	21137	BLACK RECORD COVER FRAME	11651	40633.38
17	22502	PICNIC BASKET WICKER 60 PIECES	61	39619.50
18	23284	DOORMAT KEEP CALM AND COME IN	5487	38133.64
19	22720	SET OF 3 CAKE TINS PANTRY DESIGN	7483	38108.89
20	22960	JAM MAKING SET WITH JARS	8695	37082.13

6. Data Visualization in Power BI

Finally, built an interactive Power BI dashboard to visualize key business metrics and insights.



7. Business Recommendations

- Focus marketing efforts on high-revenue products to maximize sales.
- Retain high-value customers through loyalty and personalized engagement programs.
- Optimize inventory for best-selling products to reduce stock shortages.
- Increase sales in underperforming countries through targeted marketing campaigns.
- Prepare inventory for seasonal demand, especially during peak sales months.
- Monitor both sales volume and revenue, as high-selling products do not always generate the highest revenue.